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Effects of Gravity on Interdendritic Flow: An Analytic Approach

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Interdendritic flow continues to be the subject of considerable interest by experimentalists and numerical modelers.[1,2,3] As an aid to understanding the essence of the gravitational influence on interdendritic flow, we present in this note an analytic solution of the equations governing interdendritic fluid flow for the simple case shown in Figure 1: a semi-infinite mushy zone whose isotherms are flat, parallel, and inclined at an angle to the force of gravity. The situation is modeled using the equations and boundary conditions for interdendritic flow presented by Mehrabian, Keane, and Flemings.[4] Their Eqs. [1], [2], and [3] simplify in our situation to

$$\frac{\partial}{\partial t}[(\rho_L - \rho_s)f_L] = -\frac{\partial}{\partial x_\perp}(\rho_L f_L v_\perp) \qquad [1]$$

$$\frac{\partial f_L}{\partial t} = -\left(\frac{1-\beta}{1-k}\right)\frac{f_L}{C_L}\frac{\partial C_L}{\partial T}\left[\frac{\partial T}{\partial t} + v_\perp \frac{\partial T}{\partial x_\perp}\right] \qquad [2]$$

$$\mathbf{v} = -(K/\mu f_L)[\nabla P - \rho_L \mathbf{g}] \qquad [3]$$

with the boundary condition

$$P = P_0 - \rho_L \mathbf{g} \cdot \mathbf{x} \qquad [4]$$

at the liquidus isotherm.

In [1] through [4] $(x_\parallel, x_\perp)$ represent coordinate axes, respectively, parallel to and perpendicular to isotherms. ρ_L, ρ_s, μ, β, and C_L are, respectively, the densities of liquid and solid, the liquid viscosity, the solidification shrinkage, and the liquid composition. k is the partition ratio for the solute and $\mathbf{g}$ is the acceleration due to gravity. T is the temperature and the unknowns are the volume fraction liquid f_L, the components $v_\parallel$ and $v_\perp$ of interdendritic velocity, and the pressure P. The permeability K of the interdendritic network is a function of f_L. In obtaining [1] through [4], gradients of f_L, $v_\parallel$, and $v_\perp$ parallel to the isotherms have been ignored.

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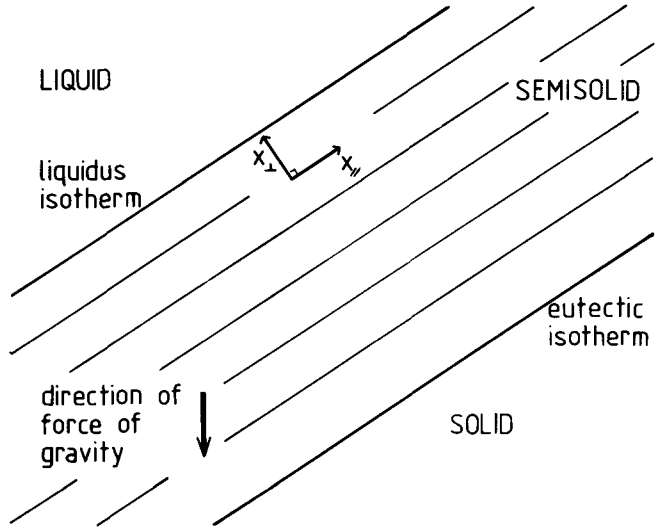

Fig. 1—Situation analyzed: flat, parallel isotherms at an angle to the vertical.

After taking derivatives of [3], one obtains

$$\frac{\partial}{\partial x_\perp}(v_\parallel \mu f_L/K) - \frac{\partial}{\partial x_\parallel}(v_\perp \mu f_L/K) = \frac{\partial}{\partial x_\perp}(\rho_L g_\parallel) \qquad [5]$$

and, after ignoring velocity derivatives with respect to $x_\parallel$ and then integrating using the boundary condition [4], [5] becomes

$$v_\parallel = \left(\frac{K g_\parallel}{\mu f_L}\right)[\rho_L(T) - \rho_L(T_L)] \qquad [6]$$

where T_L is the liquidus temperature.

Equations [1] and [2] have been analyzed by Flemings and co-workers.[5,6,7] They represent interdendritic flow induced by solidification shrinkage. It should be noted that $v_\perp$ is independent both of the force of gravity and of the permeability of the dendrite network.

Equation [6] is new. It describes a flow of interdendritic liquid parallel to isotherms. Unlike the shrinkage-induced flow, it is independent of the cooling rate. Its magnitude increases with the slope of the isotherms and with the permeability of the dendrite network.

The most important influence on $v_\parallel$ is the permeability. In Figure 2, we have plotted forms obtained by Ohnaka and Fukusako[8] from experimental measurements of the permeability K as a function of volume fraction liquid in the mushy zone. The figure shows that the permeability increases very rapidly at high volume fractions liquid, where the dendrite network is sufficiently open that the percolation process (which causes K to be proportional to f_L^2) is no longer a good description of the flow.[10] Since the permeability is far larger in the regions with a high volume fraction liquid than elsewhere, from [6], interdendritic flow should be concentrated there.

To illustrate this, we have plotted in Figure 3 $v_\parallel$ as a function of distance from the eutectic using numerical values appropriate for 10 cm thick Waspaloy mushy zone solidifying at 100 μm per second inclined at 45 deg to the vertical using data given by Jeanfils *et al.*[9] To show the way $v_\parallel$ depends on permeability, two forms for the permeability

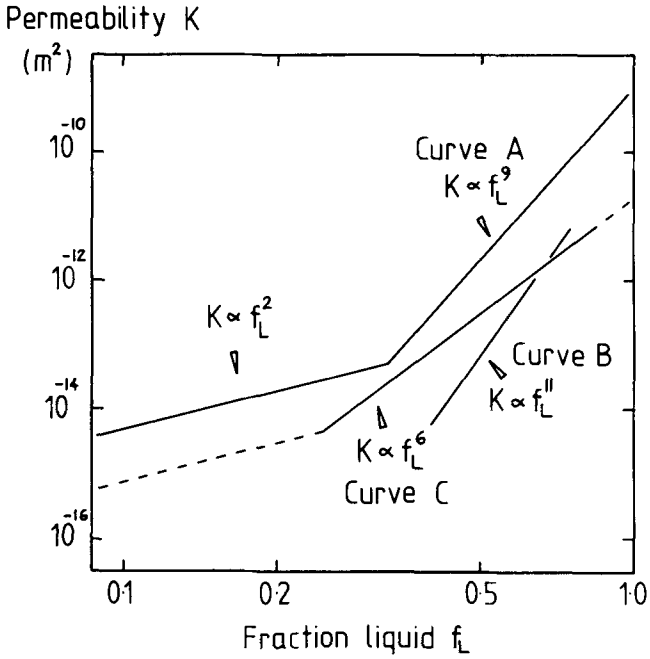

Fig. 2—Fits to experimental measurements of the permeability of an aluminum alloy[10] (curve A) and steel alloys[11,12] (curves B, C). The dashed lines indicate extrapolations made for the calculations presented in Fig. 3.

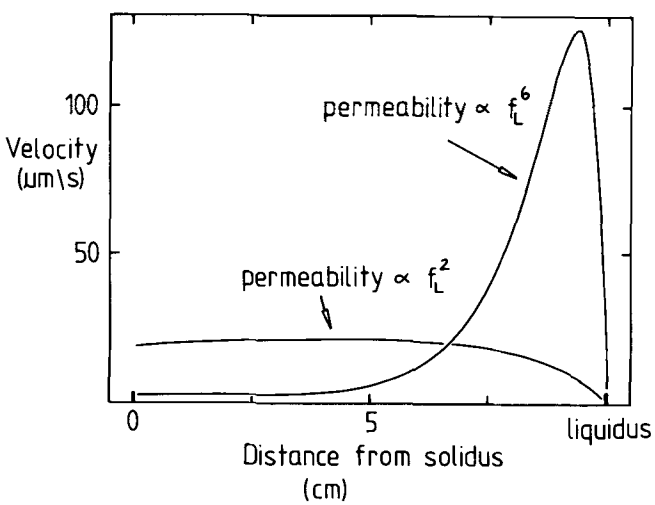

Fig. 3—Parallel component of interdendritic flow.

have been used:

$$K = a_1 f_L^2 \qquad [7]$$

$$K = \begin{cases} a_2 f_L^2 & f_L < 0.245 \\ a_3 f_L^6 & f_L > 0.245 \end{cases} \qquad [8]$$

The constants a_1, a_2, and a_3 have been adjusted to give the same net solute flow in each case.

The form [7] has been used in most numerical calculations of interdendritic flow. It agrees[10] with experimental measurements reasonably well at low fractions liquid and has been used frequently in numerical calculations to describe flow at all fractions liquid because of its simplicity. However, at high fractions liquid, Figure 2 shows that the permeability varies much more rapidly than [7] would suggest. Fluid flow parallel to isotherms according to this form is spread over the whole mushy zone.

The form [8] was chosen to fit the variation of permeability with fraction solid indicated by the curve C in Figure 2. Although the variation of permeability with fraction solid is slower for curve C than for the others, even it produces a large concentration of interdendritic flow near the liquidus parallel to it. The flow distribution differs considerably from that obtained using [7], indicating that the numerical calculations based in [7] will not generally give a reliable picture of gravity-induced interdendritic flow.

One should note that there are assumptions in the derivation of [6] which make it at best semi-quantitatively accurate. Near the liquidus, the dendrite network will not behave as a simple porous solid: some convection from the liquid above the mushy zone will penetrate into the dendrite network,[1] and viscous drag from this will either accelerate or retard interdendritic flow. If the dendrite network is weak, dendrites may break off and float freely.

We now consider how the calculations for a semi-infinite mushy zone relate to the situation in a real solidifying ingot. In several forms of casting into a water-cooled mold (*e.g.*, electroslag remelting) the mushy zone is often long and narrow.[9] It is therefore reasonable to regard these mushy zones as semi-infinite ones with flat isotherms when considering gravity-induced interdendritic flow, especially as [6] indicates that the flow tends to be confined to a neighborhood of the liquidus.

The main reason interdendritic flow is of interest is the problem of macrosegregation. In the semi-infinite mushy zone, f_L, and hence macrosegregation, is unaffected by $v_{\parallel}$, according to [1] and [2]. However, in a real ingot, flow parallel to isotherms must eventually meet the surface or the center of the ingot. In these positions, the gravity-induced flow component is diverted away from being parallel to isotherms and therefore produces segregation. The character of the segregation may be found by determining the direction of flow indicated by [6]: in a cylindrical ingot, the solute-rich interdendritic flow will go upward and hence deplete the ingot center of solute if the density of interdendritic liquid decreases with decreasing temperature (*e.g.*, Waspaloy), whereas, if it increases (*e.g.*, Ni-Mo alloys), the ingot center will be enriched.

[6] indicates that there are two ways to reduce the flow parallel to isotherms and hence segregation. First, if the isotherms are more nearly horizontal, $g_{\parallel}$ and so $v_{\parallel}$ will be reduced. Second, if the cooling rate or the thermal gradient in the mushy zone is increased, the total flux of solute will be decreased: since $v_{\parallel}$ is independent of thermal gradient, increasing the thermal gradient will reduce the width of the mushy zone and thus the total flow parallel to isotherms (although the flow of solute per unit area normal to the isotherms will be the same). Increasing the cooling rate will reduce the time available for solute flow to occur, and hence reduce the segregation.

In conclusion, we have shown how interdendritic flow can be calculated analytically in a simple case which includes the effects of gravity. The main conclusion is that gravity in ingots with narrow mushy zones mainly produces interdendritic flow parallel to the liquidus confined to the vicinity of the liquidus. This flow can be large in comparison to the usual shrinkage-induced flows. Its existence

is caused by the rapid increase of permeability with fraction liquid near the liquidus, and so it cannot appear in numerical calculations which ignore this rapid variation of permeability.

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On Yield Loci of HY80, HY100 Steels and Ti-6Al-2Nb-1Ta-0.8Mo

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It is well established that the plastic response of isotropic metals or alloys subjected to multiaxial states of stress can be characterized by the J_2-based von Mises criterion (J_2 is the second deviatoric stress invariant). For a von Mises solid, the yield function and the plastic strain ratio ρ for plane stress ($\sigma_3 = 0$) are as follows:[1]

$$\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2 = \sigma_y^2 \qquad [1]$$

$$\frac{d\varepsilon_2^p}{d\varepsilon_1^p} = \rho = \frac{2\alpha - 1}{2 - \alpha}; \qquad \alpha = \frac{\sigma_2}{\sigma_1} \qquad [2]$$

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where σ_1 and σ_2 are the principal stresses, and σ_y is the yield stress of the material under uniaxial tension. It has long been recognized that metals such as copper, aluminum, and mild steel exhibit slight deviations to Eq. [1].[2,3] The effective stress-strain curves of these materials have been found to vary with the stress state, and the apparent hardening behavior at or near the stress state of plane strain is lower when compared to that measured at either uniaxial or equibiaxial tension. In addition, the incremental plastic strain ratio at a given stress state has also been found to deviate from the von Mises predicted value. Considered as second order effects,[4] these small deviations are often neglected but can be taken into account by adding a small J_3 (the third deviatoric stress invariant) term in the yield function,[4,5,6] as in Drucker theory.[6] The Drucker yield function can be expressed as[7]

$$\left(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2\right)\left[\frac{27}{27 - 4C}\left(1 - C\frac{J_3^2}{J_2^3}\right)\right]^{1/3} = \sigma_y^2, \qquad [3]$$

and the corresponding plastic strain ratio is[7]

$$\rho = \frac{9(2\sigma_2 - \sigma_1)J_2^2 - 2C(2\sigma_2^2 - 2\sigma_1\sigma_2 - \sigma_1^2)J_3}{9(2\sigma_1 - \sigma_2)J_2^2 - 2C(2\sigma_1^2 - 2\sigma_1\sigma_2 - \sigma_2^2)J_3} \qquad [4]$$

where

$$J_2 = \frac{1}{3}(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)$$

$$J_3 = \frac{-1}{27}[(2\sigma_1 - \sigma_2)(2\sigma_2 - \sigma_1)(\sigma_1 + \sigma_2)]$$

and C is a fitting parameter. (Notice that Drucker's yield function reduces to von Mises criterion when $C = 0$.) On the other hand, the yield locus calculation based on Bishop-Hill theory[8] by Bassani[9] indicates that the deviations to von Mises criterion exhibited by bcc and fcc metals can be rationalized in terms of $\{110\}\langle 111\rangle$ and $\{111\}\langle 110\rangle$ slip, respectively. Thus, the second order effect appears to originate from the slip process and the crystallography of the cubic metals. In a recent study, Chan, Lindholm, and Wise[7] confirmed the Bishop-Hill $\{110\}$ yield locus in HY80 steel. This paper extends the earlier findings by examining the biaxial yield behavior of Ti-6Al-2Nb-1Ta-0.8Mo (Ti-6-2-1-1) and HY100 steel. The results of this study will be compared with previous findings of HY80 steel to assess the effect of crystallography on the shape of the yield surface.

The yield loci were measured by testing thin wall tubes (1.3 mm wall thickness) under combined axial tension and internal pressure. All tubular specimens were machined from commercial grade materials in the form of 30-mm thick plate and were tested in the as-received condition. The experimental setup and procedures were described elsewhere.[7,10] Figure 1(a) shows the microstructure of Ti-6-2-1-1 which consists of equiaxed primary α (hcp) and Widmanstätten $\alpha + \beta$ (bcc) pockets. Both HY80 and HY100 steels consist of a tempered martensite (ferrite plus carbide) microstructure with, occasionally, some inclusions [Figures 1(b) and 1(c)]. The grain sizes of the three materials are approximately 12 μm, resulting in a tube wall thickness to grain size ratio of 100.

Both the shape of the yield surfaces and the plastic strain ratios ($\rho = d\varepsilon_2^p/d\varepsilon_1^p$) were used to compare experimental